For program 1

```
#include <stdio.h>

#define MAX 10

/* Function Prototypes */
void readMatrix(int mat[MAX][MAX], int rows, int cols);
void addMatrix(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int rows, int cols);
void subtractMatrix(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int rows, int
cols);
void multiplyMatrix(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int r1, int c1,
int c2);
void transposeMatrix(int mat[MAX][MAX], int trans[MAX][MAX], int rows, int cols);
void displayMatrix(int mat[MAX][MAX], int rows, int cols);

int main()
{
   int A[MAX][MAX], B[MAX][MAX], C[MAX][MAX], T[MAX][MAX];
   int r1, c1, r2, c2;
   int choice;
   int matrixRead = 0;

   while (1)
   {
      printf("\n========= MATRIX MENU =========\n");
      printf("1. Read Matrices\n");
      printf("2. Addition\n");
      printf("3. Subtraction\n");
      printf("4. Multiplication\n");
      printf("5. Transpose (Matrix A)\n");
      printf("6. Display Matrices\n");
      printf("7. Exit\n");
      printf("Enter your choice: ");

      /* Parameter Safety: Validate input */
      if (scanf("%d", &choice) != 1)
      {
         printf("Invalid input!\n");
         while(getchar()!='\n');
         continue;
      }

      /* Parameter Safety: Validate menu choice */
      if (choice < 1 || choice > 7)
      {
         printf("Invalid menu choice!\n");
         continue;
      }

      switch(choice)
      {
         case 1:

            printf("Enter rows and columns of Matrix A: ");
```

```
    if(scanf("%d%d",&r1,&c1)!=2)
    {
       printf("Invalid input!\n");
       return 0;
    }

    if(r1<=0 || c1<=0 || r1>MAX || c1>MAX)
    {
       printf("Invalid dimensions for Matrix A!\n");
       break;
    }

    printf("Enter rows and columns of Matrix B: ");

    if(scanf("%d%d",&r2,&c2)!=2)
    {
       printf("Invalid input!\n");
       return 0;
    }

    if(r2<=0 || c2<=0 || r2>MAX || c2>MAX)
    {
       printf("Invalid dimensions for Matrix B!\n");
       break;
    }

    printf("\nEnter Matrix A:\n");
    readMatrix(A,r1,c1);

    printf("\nEnter Matrix B:\n");
    readMatrix(B,r2,c2);

    matrixRead=1;
    break;

 case 2:

    if(matrixRead==0)
    {
       printf("Read matrices first!\n");
       break;
    }

    if(r1!=r2 || c1!=c2)
    {
       printf("Addition not possible.\n");
       break;
    }

    addMatrix(A,B,C,r1,c1);

    printf("\nResult of Addition:\n");
    displayMatrix(C,r1,c1);

    break;
```

```
case 3:

    if(matrixRead==0)
    {
        printf("Read matrices first!\n");
        break;
    }

    if(r1!=r2 || c1!=c2)
    {
        printf("Subtraction not possible.\n");
        break;
    }

    subtractMatrix(A,B,C,r1,c1);

    printf("\nResult of Subtraction:\n");
    displayMatrix(C,r1,c1);

    break;

case 4:

    if(matrixRead==0)
    {
        printf("Read matrices first!\n");
        break;
    }

    if(c1!=r2)
    {
        printf("Multiplication not possible.\n");
        break;
    }

    multiplyMatrix(A,B,C,r1,c1,c2);

    printf("\nResult of Multiplication:\n");
    displayMatrix(C,r1,c2);

    break;

case 5:

    if(matrixRead==0)
    {
        printf("Read Matrix A first!\n");
        break;
    }

    transposeMatrix(A,T,r1,c1);

    printf("\nTranspose of Matrix A:\n");
    displayMatrix(T,c1,r1);
```

```
            break;

        case 6:

            if(matrixRead==0)
            {
                printf("Read matrices first!\n");
                break;
            }

            printf("\nMatrix A:\n");
            displayMatrix(A,r1,c1);

            printf("\nMatrix B:\n");
            displayMatrix(B,r2,c2);

            break;

        case 7:

            printf("Program Terminated.\n");
            return 0;
      }
   }

   return 0;
}

/* Function to Read Matrix */
void readMatrix(int mat[MAX][MAX], int rows, int cols)
{
   int i,j;

   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         scanf("%d",&mat[i][j]);
      }
   }
}

/* Function to Add Matrices */
void addMatrix(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int rows, int cols)
{
   int i,j;

   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         result[i][j]=a[i][j]+b[i][j];
      }
   }
```

```
}

/* Function to Subtract Matrices */
void subtractMatrix(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int rows, int
cols)
{
   int i,j;

   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         result[i][j]=a[i][j]-b[i][j];
      }
   }
}

/* Function to Multiply Matrices */
void multiplyMatrix(int a[MAX][MAX], int b[MAX][MAX], int result[MAX][MAX], int r1, int c1,
int c2)
{
   int i,j,k;

   for(i=0;i<r1;i++)
   {
      for(j=0;j<c2;j++)
      {
         result[i][j]=0;

         for(k=0;k<c1;k++)
         {
            result[i][j]+=a[i][k]*b[k][j];
         }
      }
   }
}

/* Function to Find Transpose */
void transposeMatrix(int mat[MAX][MAX], int trans[MAX][MAX], int rows, int cols)
{
   int i,j;

   for(i=0;i<rows;i++)
   {
      for(j=0;j<cols;j++)
      {
         trans[j][i]=mat[i][j];
      }
   }
}

/* Function to Display Matrix */
void displayMatrix(int mat[MAX][MAX], int rows, int cols)
{
   int i,j;
```

```
    for(i=0;i<rows;i++)
    {
        for(j=0;j<cols;j++)
        {
            printf("%4d",mat[i][j]);
        }
        printf("\n");
    }
}
```

Program-2

Program : Bank Account Management System
Concept : Parameter Passing and Parameter Safety

```
#include<stdio.h>
#include<string.h>

#define MAX_NAME 30

/* Function Prototypes */

float createAccount(int accNo, char name[], float balance);
float deposit(float balance, float amount);
float withdraw(float balance, float amount);
float checkBalance(float balance);
void displayAccount(int accNo, char name[], float balance);

int main()
{
   int choice;
   int accountNo=0;
   char customerName[MAX_NAME];
   float balance=0;
   float amount;

   int accountCreated=0;

   while(1)
   {
      printf("\n=====================================");
      printf("\n BANK ACCOUNT MANAGEMENT SYSTEM");
      printf("\n=====================================");
      printf("\n1. Create Account");
      printf("\n2. Deposit Money");
      printf("\n3. Withdraw Money");
      printf("\n4. Check Balance");
      printf("\n5. Display Account Details");
      printf("\n6. Exit");

      printf("\n\nEnter your choice : ");

      /* Parameter Safety */
      if(scanf("%d",&choice)!=1)
      {
         printf("\nInvalid input!");
         while(getchar()!='\n');
         continue;
      }

      if(choice<1 || choice>6)
```

```
{
   printf("\nInvalid Menu Choice.");
   continue;
}

switch(choice)
{

case 1:

   printf("\nEnter Account Number : ");

   if(scanf("%d",&accountNo)!=1)
   {
      printf("\nInvalid Account Number.");
      while(getchar()!='\n');
      break;
   }

   if(accountNo<=0)
   {
      printf("\nAccount Number must be positive.");
      break;
   }

   while(getchar()!='\n');

   printf("Enter Customer Name : ");
   fgets(customerName,MAX_NAME,stdin);

   customerName[strcspn(customerName,"\n")]='\0';

   if(strlen(customerName)==0)
   {
      printf("\nCustomer Name cannot be empty.");
      break;
   }

   printf("Enter Initial Balance : ");

   if(scanf("%f",&balance)!=1)
   {
      printf("\nInvalid Balance.");
      while(getchar()!='\n');
      break;
   }

   if(balance<0)
   {
      printf("\nInitial Balance cannot be negative.");
```

```
            break;
        }

        balance=createAccount(accountNo,customerName,balance);

        accountCreated=1;

        printf("\nAccount Created Successfully.\n");

        break;

    case 2:

        if(accountCreated==0)
        {
            printf("\nCreate Account First.");
            break;
        }

        printf("\nEnter Deposit Amount : ");

        if(scanf("%f",&amount)!=1)
        {
            printf("\nInvalid Amount.");
            while(getchar()!='\n');
            break;
        }

        if(amount<=0)
        {
            printf("\nDeposit Amount must be positive.");
            break;
        }

        balance=deposit(balance,amount);

        printf("\nDeposit Successful.");

        break;

    case 3:

        if(accountCreated==0)
        {
            printf("\nCreate Account First.");
            break;
        }

        printf("\nEnter Withdrawal Amount : ");
```

```
    if(scanf("%f",&amount)!=1)
    {
       printf("\nInvalid Amount.");
       while(getchar()!='\n');
       break;
    }

    if(amount<=0)
    {
       printf("\nWithdrawal Amount must be positive.");
       break;
    }

    if(amount>balance)
    {
       printf("\nInsufficient Balance.");
       break;
    }

    balance=withdraw(balance,amount);

    printf("\nWithdrawal Successful.");

    break;

case 4:

    if(accountCreated==0)
    {
       printf("\nCreate Account First.");
       break;
    }

    balance=checkBalance(balance);

    printf("\nCurrent Balance : %.2f",balance);

    break;

case 5:

    if(accountCreated==0)
    {
       printf("\nCreate Account First.");
       break;
    }

    displayAccount(accountNo,customerName,balance);

    break;
```

```
        case 6:

            printf("\nThank You...\n");
            return 0;

        }

    }

    return 0;
}

/*---------------------------------------------------*/

float createAccount(int accNo,char name[],float balance)
{
    /* Demonstrates parameter passing */

    printf("\nAccount Number : %d",accNo);
    printf("\nCustomer Name : %s",name);

    return balance;
}

/*---------------------------------------------------*/

float deposit(float balance,float amount)
{
    balance=balance+amount;
    return balance;
}

/*---------------------------------------------------*/

float withdraw(float balance,float amount)
{
    balance=balance-amount;
    return balance;
}

/*---------------------------------------------------*/

float checkBalance(float balance)
{
    return balance;
}

/*---------------------------------------------------*/
```

```
void displayAccount(int accNo,char name[],float balance)
{
   printf("\n");
   printf("\n--------- ACCOUNT DETAILS ---------");
   printf("\nAccount Number : %d",accNo);
   printf("\nCustomer Name  : %s",name);
   printf("\nCurrent Balance: %.2f",balance);
   printf("\n-----------------------------------");
}
```